

Reg. No.: _____

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B.Tech Mid Semester Examination – September 2025

Seventh Semester CSE

Computer Science and Engineering

19CSE353 Mining of Massive Datasets

Duration : Two hours

Maximum : 50 Marks

CO	Course Outcomes
CO1	Understand the importance of data to business decisions, strategy and behavior. Predictive analytics, data mining and machine learning as tools give new methods for analyzing massive data sets.
CO2	Expose the students to big data systems like Hadoop, Spark and Hive.
CO3	Explore means to deal with huge document databases and infinite streams of data to mining large social networks and web graphs. Also, learn Algorithms suitable for large scale mining.
CO4	Use of analytic tools, case-studies to provide firsthand insight into how big-data problems and their solutions are commercially derived.
CO5	Design Large Scale Machine Learning algorithms with Practical hands-on experience for analyzing very large amounts of data.

Answer all the questions

1. a Assume a repository consists of $2^{18} = 262,144$ documents. A certain word [4] [CO1]
"machine" appears in $2^8 = 256$ of these documents. [BTL3]
 - i. Compute the IDF of the word machine using base-2 logarithm.
 - ii. Consider a document A in which the word machine appears 15 times, and this is the maximum frequency of any word in that document. Compute the TF and the TF-IDF score for machine in document A.
 - iii. In another document B, the word machine appears 3 times, while the maximum word frequency in document B is 30. Compute the TF and the TF-IDF score for machine in document B.
 - iv. Based on your calculations, in which document (A or B) is the word machine given more importance?
- b A healthcare research organization is analyzing a massive dataset of clinical [2+1] [CO1]
reports and patient medical histories to identify disease patterns and improve [BTL2]
diagnosis. The dataset contains long textual descriptions, lab results, and doctor notes for millions of patients. Discuss how summarization and feature extraction complement each other in making healthcare data more useful for predictive analytics and machine learning applications.
- c An e-commerce company integrates browsing history, transaction records, [2+1] [CO1]
location data, and social media reviews into its big data system for fraud detection [BTL2]
and recommendations.

- i. Explain how the company's data integration approach reflects the concept of Total Information Awareness (TIA). [2+2+1] [CO2] [BTL3]
- ii. State one benefit and one risk in this context.

2. a A teacher wants to count how many times each word appears in her students' short essays using MapReduce. Each essay is treated as a document (an input element). The Map function outputs (word, 1) for every word in the essay. The system then groups by key (word) and sends the grouped pairs to reducers. The Reduce function adds up the counts for each word. [2+2+1] [CO2] [BTL3]

Suppose there are two essays:

Essay 1: "AI is fun"

Essay 2: "AI and data"

- i. Write the Map output for each essay.
 - ii. Show the grouped key-value pairs before reduction.
 - iii. Write the final Reduce output.
- b A university is analyzing network traffic between two departments. The traffic is represented as an adjacency matrix M , where each row corresponds to a computer (sender) and each column corresponds to a computer (receiver). A load vector v , represents the traffic weight to each receiver. [2+2+1] [CO1] [BTL3]

$$M = \begin{bmatrix} 1 & 0 \\ 4 & 2 \end{bmatrix}, \quad v = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$$

- i. Explain how this computation can be performed using a two-phase MapReduce approach (mapper and reducer).
 - ii. Show the intermediate key-value pairs emitted by the mapper.
 - iii. Show the reducer outputs (final engagement scores for each student).
3. A university is building a plagiarism detection tool using character shingles. Each document is broken into k-character shingles to compare overlaps. Consider two student submissions: [2+2+1] [CO2] [BTL3]

Doc A: "Data mining techniques"

Doc B: "Data analysis techniques"

If $k = 4$,

- i. Generate the set of shingles for each document.
- ii. Identify the overlapping shingles between the two documents.
- iii. Explain why character-level shingles may capture similarity better than word-level shingles in this case.

4. Explain the concept of standing and ad-hoc queries used in the stream processing model with an example. [4] [CO3] [BTL2]

- a. Identify the approximate number of 1's in the infinite stream 1, 1, 0, 0, 1, 1, 0, 0, [3] [CO3]

1, 0 assuming the window size as 10. [BTL3]

- b. If the stream of inputs arriving at a device is [12, 7, 15, 7, 9, 12, 18, 15, 21, 7, 9, 18], calculate the total number of distinct elements in it using Flajolet-Martin algorithm. Use the Hash function $h(x) = (3x + 5) \bmod 32$ (4-bit binary). [CO3] [BTL3]
- c. You are storing elements in a Bloom filter with an array numbered from 0 to 14. The stream that arrives for storage is 23, 51, 36, 92, 47, 19, 64, 87, 41, 75. You have decided to use the hash function $h(x) = (2x + 4) \bmod 15$. Show how the Bloom filter works in storing these elements and how does false positives occur in the process. [CO3] [BTL3]
6. The database given below in Table 2 has four transactions from four customers. Using the Apriori algorithm, find the association rules in this set, if the minimum support (i.e., coverage) is 30% and the minimum confidence (i.e., accuracy) is 50%. [Note: minimum support threshold is 30% of the total transactions]. [CO3] [BTL3]

Table 2

Customer	Items
1	Bread, Milk, Butter
2	Bread, Milk, Eggs
3	Bread, Butter, Jam
4	Milk, Butter, Eggs

7. Consider the web graph shown in Figure 1. Calculate the PageRank values for each page using two iterations of the PageRank algorithm, assuming a damping factor of 0.85. Also, explain how the damping factor influences the ranking distribution. [CO3] [BTL3]

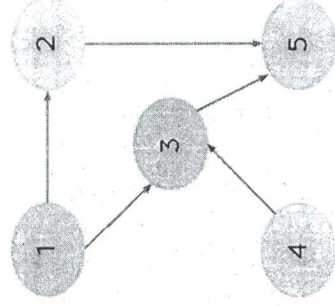


Figure 1

Course Outcome / Bloom's Taxonomy Level (BTL) Mark Distribution Table

CO	Marks	BTL	Marks
CO1	15	BTL 1	0
CO2	10	BTL 2	10
CO3	25	BTL 3	40
CO4	0	BTL 4	0
CO5	0	BTL 5	0
		BTL 6	0